Lecture 10 Team Projects

Credit Spread Exposure and Credit Risk Modeling

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1 Project 1: Spread Duration and Credit Scenario Budget

Scenario. Determine how much spread widening would erase one year of portfolio carry.

Objectives. Estimate carry and spread duration; calculate break-even widening; distinguish spread, default, Treasury-rate, and liquidity risk.

Sources. FINRA TRACE and issuer filings; FRED OAS indexes; U.S. Treasury data; instructor analytics.

Deliverables. Position profile, calculations, break-even result, three scenarios, risk-budget recommendation, and a 10–12 minute presentation.

Rubric (100 points). Inputs 20; calculations 30; scenarios 20; decomposition and recommendation 20; presentation 10.

2 Project 2: Expected Loss and Bondholder Recovery

Scenario. Compare two speculative-grade bonds with different default probabilities, seniority, collateral, and prices.

Objectives. Estimate expected loss; analyze recovery drivers; compare expected loss with observed spread while recognizing risk and liquidity premiums.

Sources. SEC EDGAR or SEDAR+; FINRA; public rating-agency studies; FRED high-yield spreads; instructor default tables.

Deliverables. Capital structure, PD/recovery assumptions, loss scenarios, spread comparison, creditor preference, and a 10–12 minute presentation.

Rubric (100 points). Research 20; loss calculations 25; recovery reasoning 25; judgment 20; presentation and citations 10.

3 Project 3: Rating Migration Stress Test

Scenario. Estimate the effect of a recession-driven wave of downgrades on an investment-grade portfolio.

Objectives. Map holdings to rating buckets; apply a transition matrix; translate migrations into spread shocks and value changes; identify fallen-angel risk.

Sources. Public rating-agency transition studies; ETF holdings; FRED spread indexes; instructor spread-by-rating assumptions.

Deliverables. Starting profile, migration matrix, base and recession losses, concentration findings, action plan, and a 10–12 minute presentation.

Rubric (100 points). Portfolio mapping 20; migration analysis 25; valuation effects 25; action plan 20; presentation and reproducibility 10.

4 Project 4: Covenant and Capital-Structure Relative Value

Scenario. Choose between two bonds from one issuer that differ in seniority, security, covenants, maturity, and spread.

Objectives. Build a waterfall; compare covenant protection; estimate recovery ranges; decide whether incremental spread compensates for structural subordination.

Sources. Issuer filings and prospectuses; FINRA data; rating-agency recovery studies; instructor enterprise-value scenarios.

Deliverables. Capital-structure diagram, covenant comparison, recovery waterfall, spread-versus-risk conclusion, and a 10–12 minute presentation.

Rubric (100 points). Document research 25; waterfall 25; covenant and recovery reasoning 25; recommendation 15; presentation and sourcing 10.